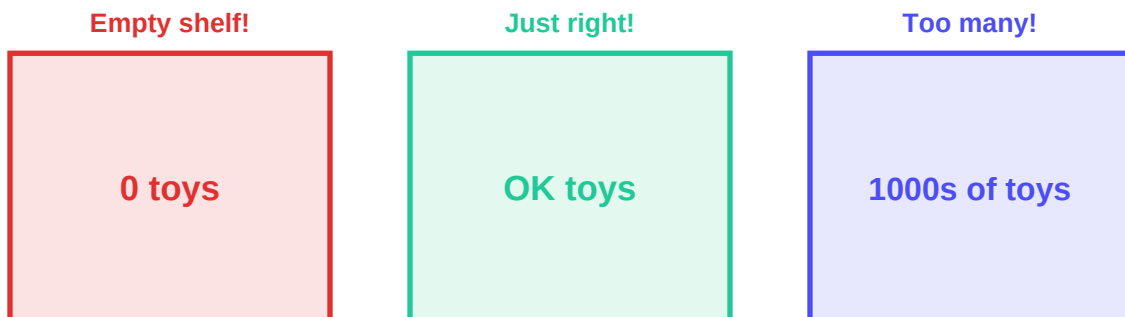


The Story of

Project FORESIGHT

*A magic helper that tells a toy shop what to order,
and what to put on sale.*



Once Upon a Time...

There was a shop called **NorthBay Living**. They sold nice things for people's homes — chairs, lamps, blankets, and more.

But the shop had a problem. They didn't know how many of each thing to buy!

Problem 1: Empty Shelves

Sometimes a popular thing (like a comfy chair) sold out. People wanted to buy it, but there was none left! The shop lost money it could have made.

Problem 2: Too Many Things

Sometimes the shop bought WAY too much of something nobody wanted. It just sat there taking up space, and the money spent on it was stuck.

So the shop asked for help: **“Can you tell us what to buy, and when?”** That's the job I was given — and this book tells you how I solved it, step by step, like a recipe.

The Six Steps

I solved the shop's problem in six steps, like following a recipe. Each step builds on the one before it.

1 Tidy Up the Messy Notes
Clean the shop's messy sales records.

2 Look for Clues
Find patterns — what sells, what doesn't.

3 Guess the Future
Predict how much will sell next.

4 Traffic Lights
Decide what to buy more of, and what to clear.

5 Build a Magic Screen
A simple tool anyone can use.

6 Tell the Grown-Ups
Explain it all in a simple slideshow.

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Step 1: Tidy Up the Messy Notes

Imagine the shop's sales notes are like a messy toy box. Some notes are written twice by mistake. Some are missing pieces. Some spell things differently (like “Furniture” and “furniture” — that's the same thing, just written differently!).

Before I could learn anything from the notes, I had to **tidy them up first**:

- Threw away notes that were exact copies (duplicates)
- Filled in missing pieces with a smart, fair guess
- Made sure everything was spelled the same way

Tool used: Python (pandas)

Think of Python as a robot helper, and pandas as its favorite toy — a giant, super-fast spreadsheet that can clean thousands of rows in seconds.

Step 2: Look for Clues

Now that the notes were tidy, I played detective! I looked through all the clean sales notes to find clues about what people like to buy.

Here's what I found:

Some things are superstars

A few products sold TONS. A few products barely sold at all (we call these 'sleepy toys' — dead stock).

Winter is the best season

People bought about 30% more in Winter than in Summer. The shop should stock up before Winter!

Sale days really work

When something was on sale, people bought about 50% more of it. Sale days are a real, useful clue — not just luck.

Tool used: Charts (matplotlib)

I drew simple pictures (bar charts) so anyone — even a grown-up who doesn't code — could see the clues at a glance.

Step 3: Guess the Future

This was the hardest and most important step: **predicting how much of each thing will sell next week.**

But here's an important rule I followed: **never trust a fancy guess until it beats a simple guess first.**

The Simple Guess (the baseline)

"This week will probably look like the same week last year." That's it — no computer brain needed, just copying last year. This has to be the easiest bar to beat.

The Smart Guess (the model)

A computer program that learns from lots of clues at once: recent sales, the season, whether there's a sale on, and more — all at the same time.

Who won?

	How wrong was the guess? (smaller = better)
Simple Guess	21.9%
Smart Guess	20.5% - Winner!

The Smart Guess won — but only a little bit. I could have pretended it was way better, but that wouldn't be honest. Telling the truth about how good (or not perfect) something is matters a lot in real work.

Step 4: Traffic Lights for Products

Knowing the future isn't enough — the shop needs to know **what to DO about it**. So I gave every product a traffic-light-style label:

Healthy (Green)	This one is just fine. Leave it alone.
Reorder Now (Red)	This will run out soon! Buy more.
Clear Stock (Blue)	There's way too much. Put it on sale.
Watch Closely (Purple)	Something odd is happening — keep an eye on it.

What we found when we checked all 203 products:

109

products are Healthy

71

products need Reordering — about \$1.74 million could be lost if ignored!

23

products should go on sale — about \$539 thousand is stuck in them

Step 5: The Magic Screen

All this clever thinking lives inside computer code — but the shop's team doesn't know how to code! So I built them a **magic screen** (called a “dashboard”) that anyone can use, just by clicking buttons.

On this screen, they can:

- Pick which type of product to look at (like just “Chairs” or just “Blankets”)
- See a colorful picture of which products are risky
- See a simple list: “Buy these next”, sorted by what matters most

Tool used: Streamlit

Streamlit turns plain Python code into a real website — no need to know web design at all. And it lives on the internet, so anyone with the link can open it, anytime, from any device.

Step 6: Tell the Grown-Ups

The people running the shop are busy and don't want to read code or charts full of confusing words. So the last step was making a simple slideshow just for them.

The golden rule for this slideshow:

Lead with the money

Grown-ups in charge of a business care most about money. So the very first thing the slideshow says is: "Here's \$1.74 million you could lose, and \$539 thousand that's stuck — here's what to do about it." Then it explains everything else.

It also honestly explains what the tool CAN'T do yet — like it can't guess perfectly for a brand-new product with no sales history. Being honest about weaknesses makes people trust you more, not less.

My Toolbox

Here's every tool I used, explained super simply:

Python	The robot helper that follows my instructions, step by step.
pandas	A giant, super-fast spreadsheet Python uses to clean and sort data.
matplotlib	Draws simple charts and pictures from numbers.
scikit-learn	A toolbox of “learning” tricks — this is what powered the Smart Guess.
Jupyter Notebook	A notebook where I write code and see results right away, step by step.
Streamlit	Turns Python code into a real, clickable website — the Magic Screen.
GitHub	A safe online box that stores all my work and its history.
PowerPoint Deck	Where I explained everything to the grown-ups, simply.

The End!

So that's the whole story: a shop had a problem, and step by step — tidy the notes, look for clues, guess the future honestly, sort things with traffic lights, build a magic screen, and explain it simply — the problem got solved.

And the best part? Every single step can be explained to **anyone** — even a small kid — because good ideas don't need big, confusing words.